

6. If you click 'switch-on' button, you can see live video being captured

by the camera on your PC screen.

Note that, since the processor

speed is limited to 40 MHz, we cannot increase camera speed baud rate beyond 57600. Hence we cannot display a smooth live video.

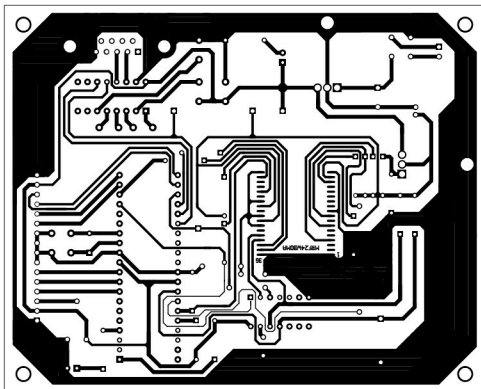


Fig. 8: Actual-size, single-side PCB layout of the Wi-Fi embedded system

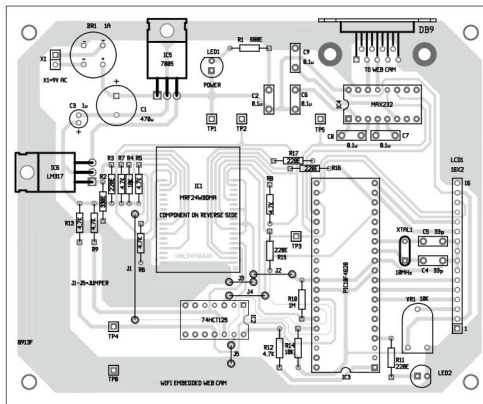


Fig. 9: Component layout for the PCB



Fig. 10: Main web page



Fig. 11: Captured video

You can click 'switch-off' button to close the captured image.

Further applications

By careful PORT forwarding of wireless router to local IP address, appliances can be controlled through the webcam from any place in the world. Besides, by using high-speed ARM processors we can increase the camera's baud rate and thereby display live pictures smoothly by using AJAX in webpage design.

Monitoring of parameters such as temperature, pressure, voltage and current can be done via the Internet from anywhere in the world by connecting suitable sensors to input port of the microcontroller. Since PIC18F4620 has limited memory space, this feature was not included in this article. ●

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